



DEBARK UNIVERSITY MODEL EXIT EXAM FROM ‘MOSHE’

1. Which one of the following is asynchronous in message communication?

- A. Blocking receive
- B. Blocking send
- C. Direct message
- D. Non-block receive

2. What is the purpose of an opcode that is as indicated in the figure below?

bits4 bitsbitsOpcodeOpertercno

- A. It determines the operation to be done
- B. It holds the next instruction reference
- C. It holds the destination address of an operand
- D. It holds the source address of an operand

3. Which layer uses port number to identify applications?

- A. Transport layer
- B. Application layer
- C. Network layer
- D. Physical layer

4. Which asymptotic notation describes average case analysis?

A. $f(n)$ - Big-Oh

B. $f(n)=O(\theta)$

C. $f(n) = \Omega$ (Big omega)

D. $f(n) = \omega$ (little omega)

5. At which layer of the OSI model devices such as bridges, switches and Network Interface cards are used?

A. Physical layer

B. Network layer

C. Data link layer

D. Application layer

6. A computer expert would like to know the number of processes that are completed per hour to assess the performance of a machine. Therefore, which one of the following best describes the expert's assessment?

A. Turnaround time

B. System throughput

C. Waiting time

D. Response time

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B. System throughput

C. Waiting time

D. Response time

8. What will be the output of the following JavaScript statement? `Math.sqrt(36)`

A. 6

B. 3

C. 36

D. 1

9. Which one of the following is correct about human role in system administration?

A. Ethics is not a requirement to manage computers

B. It doesn't need organizational skills

C. It requires patience, understanding and knowledge

D. System administrator is not expected to have skills and confidence

10. Which one of the following is valid statement regarding PHP variables?

A. `$test = "This is 'Computer science'";`

B. `test= 5;`

C. `$test= 9.5`

D. `$test = "Computer;`

11. Which one of the following is not correct about POP3 and SMTP?

A. Both are application layer protocols

B. A sender can send email using SMTP

C. A sender can receive emails using POP3

D. Both are network layer protocols

12. Which one of the following is correct for the statement below? "A problem that occurs when one transaction updates a database item and then the transaction fails for some reason and the updated item is accessed by another transaction before it is changed back to its original value"

A. It is a lost update problem

B. It is an incorrect summary problem

C. It is dirty read

D. It is a temporary read problem

13. Which quantifier can be used for some portion of the universe?

A. $\rightarrow$

B. $\exists$

C. $\wedge$

D. $\forall$

14. What will be the output of the following JavaScript statement? `Math.floor(5.9)`

A. 9

B. 5

C. 11.8

D. 6

15. Which one of the following best explains attackers who are motivated by financial gain and perform sabotage, an IP theft and secrets of a rival company?

A. Stealer

B. Competitor

C. Activist

D. Hacktivist

16. _____ is the whole thing that an agent has observed so far concerning the existing scenario in the environment.

A. Performance measure

B. Knowledge

C. Action

D. Perception

17. Which one of the following phases is done before semantic analysis phase of compiling process?

A. Syntax analysis

B. Code optimization

C. Code generation

D. Intermediate code generation

18. Which statement is false about universal hashing technique?

A. Single fixed hash function technique is better than universal hashing technique in terms of collision reduction

B. Universal hashing reduces the chance of data collision

C. In universal hashing a hash function is selected from set of hash functions independent of keys.

D. In universal hashing a hash function is selected randomly from set of hash functions for each key.

19. Which one of the following is true about functions in JavaScript?

A. Function names can contain spaces

B. Function names can begin with digits

C. Function names are not case-sensitive

D. There is no limit to the number of function parameters that a function may contain

20. Which phase of software development process refers to modification of software to reflect changing customer and market requirements?

A. Validation

B. Development

C. Specification

D. Evolution

21. Which statement is true about pop() operation of Stack data structures?

A. pop () inserts new element at the beginning/bottom of stack

B. pop () removes the end/top element from stack

C. pop () returns the end/top element but it doesn't delete that element

D. pop () inserts new element at the middle of stack

22. Which statement is false about programming language generations?

A. Assembly language is machine independent

B. Assembly language should be translated to machine language before execution

C. Machine language can be executed without translation

D. Fifth generation programming languages need very powerful hardware and software

23. Which one of the following is responsible for separating some machines from the rest to shield them from possible attacks?

A. Switch

B. Firewall

C. Router

D. Network operating system

24. Which one of the following is a two-level cache having an internal and external caches?

A. Multilevel cache

B. Single level cache

C. Unified cache

D. Split cache

25. Assume there is a line of patients waiting for a service from a clinic. Seriously sick patients should get service first. Which data structure is the most appropriate for the given type of real-world problem?

A. Stack

B. Priority Queue

C. Queue

D. Tree

26. Which one of the following is not correct about computer architecture and organization?

A. An organization can last for long time as an architecture does

B. An architecture can be used for long time encompassing different computer models

C. An architecture can remain for long time but its organization can change

D. Manufacturers offer computer with same architecture but different organization

27. Assume you have a weighted graph G with a set of vertices V and set of edges E . Which one of the following problems is an optimization problem on the graph?

A. Is vertex $p \in V$ reachable from vertex $q \in V$ in graph G

B. Find minimum spanning tree from graph G

C. Traverse all vertices in graph G

D. How many cycles are there in graph G

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29. Which one of the following is not correct about transaction failure?

A. Catastrophes can be cause of transaction failure

B. Transactions fail if disk blocks lose their data

C. Concurrency control enforcement avoids transaction failure

D. Transactions fail if logical errors are detected

30. Which one of the following is not correct about packet and circuit switching?

A. Packet switching is more efficient than circuit switching

B. In circuit switching a channel is dedicatedly used

C. In Packet switching and end-to-end connection has to be established

D. In packets switching messages are sent in small blocks

31. Which one of the following is not an example of scripting languages?

A. PHP

B. Perl

C. C++

D. Python

32. In a tree data structure, if a node has no parent node, then the node is_____

A. Root node

B. Parent node

C. Internal node

D. External node

33. _____ is a network architecture characteristic that states a network can expand quickly to address needs of new users and application without having impact on the services delivered to existing users.

A. Fault tolerance

B. Quality of service

C. Scalability

D. Security

34. Which deadlock prevention mechanisms don't require timestamp?

A. Wound-wait

B. Wait-die

C. No-wait

D. Wait-wait

35. Which one of the following lists of environments is the hardest for an agent?

A. Sequential, non-deterministic, dynamic

B. Partially observable, static, continuous

C. Fully observable, sequential, deterministic

D. Deterministic, fully observable, static

36. What does the program data independence in the database approach entails for?

A. The application may be affected when data changes

B. Data and applications are defined separately

C. Both program and data are defined together

D. Programs can create redundant data independently

37. What is the output of the following Java fragment code?

```
int n=6, m=15, p=3;
```

```
n-=m
```

```
n*=p;
```

```
System.out.print(n);
```

A. -27

B. 27

C. 3

D. 24

38. What is the time complexity order of Quick sort algorithm?

A. $O(n^2)$

B. $O(2^n)$

C. $O(n)$

D. $O(n \log_2 n)$

39. Which one of the following is the smallest heading tag?

A. <H3>

B. <H1>

C. <H4>

D. <H6>

40. If you plan a software test to be done by clients to demonstrate that the system satisfies their requirements, which phase of software testing is appropriate?

- A. Unit testing
- B. Integration testing
- C. System testing

D. Acceptance testing

41. Which one of the following is a measure taken, to ensure information and information systems' availability, integrity, authentication and confidentiality?

- A. Interception
- B. Information assurance

C. Information security

D. Wiretapping

42. Which network type is the largest as compared to the rest?

A. Metropolitan Area Network

B. The internet

C. Wide Area Network

D. Local Area Network

43. For an input graph, Depth First Traversal algorithm uses data structure to traverse?

- A. Tree
- B. Priority queue
- C. Queue

D. Stack

44. Which branch of study deals with whether a problem can be solved at all or not, regardless of the resources required?

A. Automata theory

B. Set theory

C. Computability theory

D. Complexity theory

45. Which one of the following algorithms can be used to extract Minimum Spanning Tree from an input graph?

A. Huffman encoding algorithm

B. Prim's algorithm

C. Merge sort algorithm

D. Dijkstra's algorithm

46. "A university owns a number of private local area and wide area networks, which are designed for an access by its academic staff, students and administrative workers. Which term best describes the University's network?"

A. Metropolitan Area Network

B. Local Area Network

C. Wide Area Network

D. The Internet

47. Which one of the following is not a single level ordered index?

A. Multilevel index

B. Primary index

C. Clustering index

D. Secondary index

48."Given the schemas:BOOK (Book_id, Title, Publisher name)
BOOK_AUTHOR (Book_id, Author _name) PUBLISHER (Name, Address,
Phone) Which one of the following SOL is correct to delete the publisher named as
'xWZ from the PUBLISHER table?

A. DELETE FROM PUBLISHER WHERE Address - XWZ

B. DELETE FROM BOOK WHERE Name = XWZ

C. DELETE FROM BOOK_AUTHOR WHERE Address = xWZ

D. DELETE FROM PUBLISHER WHERE Name =XWZ

49.Which one is a method used to obtain the meaning of encrypted information
without access to secret key?

A. Decryption

B. Crypto-analysis

C. Encryption

D. Cryptography

50.Which one of the following is used for the purpose of syntax analysis?

A. Loader

B. Scanner

C. Linker

D. Parser

51. Suppose you have received a message from your friend but the message is
scrambled for the sake of security. Your friend has used a program to encode
messages into scrambled text, In, order to read the message which program would
you use?

A. Secret key

B. Encryption program

C. Ciphertext

D. Decryption program

52. Which one of the following computers can be most secured compared to the rest?

A. A computer connected to the Internet and the latest anti-virus installed on it

B. A computer connected to the Internet having strong intrusion detection

C. A computer connected to the Internet with the latest firewall system

D. A computer that is not connected to the Internet

53. Which one of the following is not true about fan trap in Entity Relationship Diagram?

A. It can be resolved by readjusting the model so that no one-to-many relationships fanning out from an entity

B. When there is ambiguous pathway between some entity occurrences in related entity types

C. It occurs when there is no pathway between related entities

D. Highly likely exist when more than two one-to-many relationships fan out from an entity

54. Which one of the following is false about arrays in C++ ?

A. An array is a collection of similar data objects

B. Size of an array should be constant at the time of array declaration

C. We use [] square bracket at the time of array declaration

D. We can access elements of arrays without using index number

55. In which situation that using database management system (DBMS) becomes costly and not recommended to use it?

A. To control redundancy

B. In embedded systems having too small storage space

C. In situation where more than one users should access the database

D. To deny access to unauthorized users

56."Which one of data updates best describes the following statement? " All modified data items in the cache is written either after transaction ends its execution or after a pre-determined number of transactions have concluded their execution

A. In-place update

B. Immediate update

C. Shadow update

D. Deferred update

57.Which one of the following is true about java and JavaScript?

A. JavaScript can be used in extremely sophisticated applications

B. JavaScript is compiled and interpreted in source code form

C. Both are object-oriented languages

D. There are more features in JavaScript than Java

58.Which one of the following is a step-in problem solving that lists the actions and states that the agent has to consider provided a goal?

A. Search

B. Problem formulation

C. Execute

D. Goal formulation

59. In set of attributes that can be seen by a program and can have direct impact on the logical execution of a program.

A. Arrangement

B. Architecture

C. Organization

D. Flow control

60. Which type of Turing machine has two tapes, one tape read-only and the other read-write tape?

N.B An "Offline Turing Machine" has two tapes. One tape is read-only and contains the input, the other is read-write and is initially blank. "Offline Turing machines are equivalent to Standard Turing machines"

A. Multi-dimensional Turing machine

B. Off-line Turing machine

C. Non-deterministic Turing machine

D. Multi-head Turing machine

61. Assume you make modification to your software. Then you want to make sure that the modification has not had introduced new errors. Which software testing level is required?

A. Regression testing

B. Dynamic testing

C. Beta testing

D. Static testing

62. Which one of the following does not show non-functional requirement of software systems?

A. Displaying information

B. Memory requirement

C. Reliability

D. Response time

63. If you want to allow subclasses to access data fields or methods defined in the super-class, but not to allow non-subclasses to access these data fields and methods, which access modifier is appropriate?

A. Private

B. Default

C. Public

D. Protected

64. Which one of the following registers hold the address of the next pair of instructions to be fetched from memory?

A. Memory buffer register

B. Instruction buffer register

C. Memory address register

D. Program counter

65. Linked list data structure is _____ type of data structure.

A. Dynamic data structure

B. Non-linear data structure

C. Static data structure

D. Fixed size data structure

66. An interface that has multiple lines connecting input/output module and peripheral device and assures multiple bits to be transferred at the same time is:

A. Serial interface

B. Parallel interface

C. Bus interface

D. One line

67. Look at the following PHP line: \$list – array (20, 12, 23, 19, 18): What is the index of data value 19?

A.3

B.2

C.1

D.0

68.Which phase of compiling process is considered as the final phase in the process?

A. Code optimization

B. Code generation

C. Semantic analysis

D. Intermediate code generation

69.Which one of the following is not correct about threads?

A. Treads have program counters

B. It is a group of processes

C. Treads have registers to hold its working memory

D. Treads are the entities schedule for execution on the CPU

70.Which one of the following is false about project planning stage?

A. Project planning is a one-time task in software development life cycle

B. Cost estimation is done during project planning

C. Preparing time schedule is done during project planning

D. Risk analysis is done during project planning

71. Which type of variable can be accessed throughout our C++ program scope?

A. Static

B. Local

C. Global

D. Automatic

72. Which concept refers to extending specialized classes from generalized classes in object-oriented programming?

A. Inheritance

B. Overriding

C. Overloading

D. Data field Encapsulation

73. Suppose you have developed a self-driving car equipped with relevant sensors and camera that helps it to see the road and drive sensing traffic lights, pedestrians and signs. Its sensors are able to detect the environment but not all the aspects. Hence, in which environment is your car acting?

A. Fully observable

B. Discrete

C. Partially observable

D. Stochastic

74. Which one of the following is not correct for the figure shown in Question 27?

A. P0 is waiting for a resource held by P1

B. P1 is waiting for a resource held by P2

C. P2 cannot be executed until P1 finishes

D. B cannot be executed until P4 finishes

75. Which one of the following is correct way creating String object?

A. String m = 'Computer Science'

B. String m = 'H' 'e, ', 'I, o';

C. String m = new String ('Technology');

D. String m = new String ("Exit Exam");

76. Which one of the following is true about frames in HTML?

A. Frames allow parts of the page to remain stationary while other's parts scroll

B. All browsers support frames

C. Load on the server is not affected, if there are a large number of frames in a page

D. Frames are not difficult to handle for search engines

77. Which one of the following is false about String in Java?

A. We can create a String object using array of characters

B. Content of a string can be changed once the string is created

C. We can create a String object using String literal

D. In Java a string is treated as object

78. A database management system (DBMS) has _____ to control locks.

A. Query Processor

B. Lock Manager

C. Lock Table

D. Query Optimizer

79. Which one of the following is evaluating the degrees of success of an agent?

A. Action

B. Perception

C. Performance measure

D. Knowledge

80. Which one of the following is a security service enforced to protect disclosure of information, which can be stored in file or being on transmission, from unauthorized entities?

A. Integrity

B. Confidentiality

C. Availability

D. Authentication

81. Assume you have a universe of Key U. Which one is false statement regarding Direct Address table and Hash Table data structures?

A. Direct Address Table doesn't use hash function to map keys

B. If universe U is very large, Hash table is better than Direct Address table

C. Hash table allocates one separate memory slot for each key in the universe U

D. Here is one to one correspondence between keys in the universe U and memory slots in the Direct Address table

82. Which network security issue shows the level of weakness in networks and devices?

A. Availability

B. Vulnerability

C. Attack

D. Treat

83. What is the time complexity order of binary searching algorithm?

A. $O(1)$

B. $O(n)$

C. $O(n^3)$

D. $O(\log_2 n)$

84. After deep analysis if your software requires major change to the architecture of the system, which strategy of software change should be applied?

A. Architectural transformation

B. Preventative maintenance

C. Software reengineering

D. Software maintenance

85. Which type of variable can be accessed throughout our C++ program scope?

A. Static

B. Local

C. Global

D. Automatic

86. Which one of the following is the correct way of creating a String object?

A. `String m = 'Computer Science'`

B. `String m = 'H' 'e', '"', 'l', 'o';`

C. `String m = new String ('Technology');`

D. `String m = new String ("Exit Exam");`

87. Suppose you are administering a network and would like to know whether a computer in the upstairs is connected to the network or not. Therefore, which command would you use?

A. Traceroute

B. Ping

C. DHCP Discover

D. Ns lookup

88. Let a Grammar $G((S), \{a, b\}, S, P)$, where $\{S\}$ is set of variables, $\{a, b\}$ is set of terminal symbols, S is start variable and P is set of productions.

Then which one of the following is not an element of the language L (L) which is generated from G . Given:

$P: S \rightarrow aSb$.

$S \rightarrow \Lambda$

A. a

B. $aabb$

C. ab

D. Λ

89. Size of large data files can be reduced to save storage disk space, which algorithm is appropriate to reduce size of large files?

A. Merge sort algorithm

B. Heap sort algorithm

C. Prim's algorithm

D. Huffman encoding algorithm

90. Which asymptotic notation describes worst case analysis?

A. $f(n)$ little-oh

B. $f(n)$ - little- ω (little omega)

C. $f(n) = \theta$ (theta)

D. $f(n) = n$ (Big omega)

91. Which one of the following is different from the other based on its role in computer network?

A. Wireless access point

B. Network printer

C. Routers

D. Hubs

92. Advancement of operating system is closely tied to computer architectures, therefore, has passed through different generations. Which one of the following generation and advancement is wrongly coupled?

A. Fourth generation -> Real time systems

B. First generation -> Single user

C. Second generation > Multiprogramming

D. Third generation -> Batch system

93. Among the following, which one best describes a database management system (DBMS)?

A. It can help users to manage computer's activity

B. It can assist process images

C. It is responsible for creating and maintaining a database

D. It can assist you to process document

94. A process is terminated by the operating system due to a reference to non-existing memory. Which one of the following best explains the process termination?

A. Fatal exit

B. Normal exit

C. Error exit

D. Terminated by another process

95. Which one of the following is correct for the statement below? When two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect"

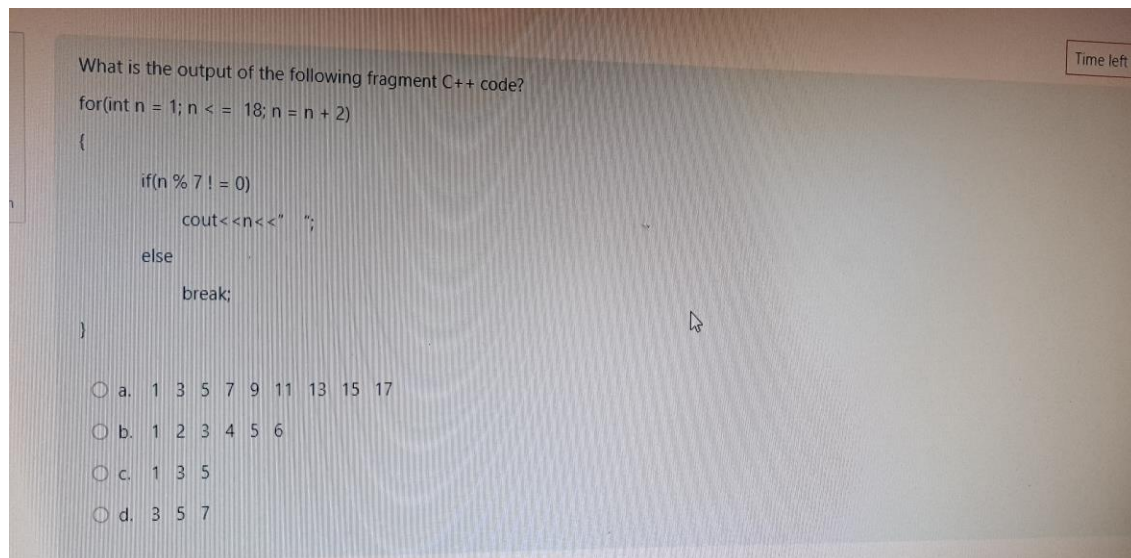
A. It is a dirty read problem

B. It is a lost update problem

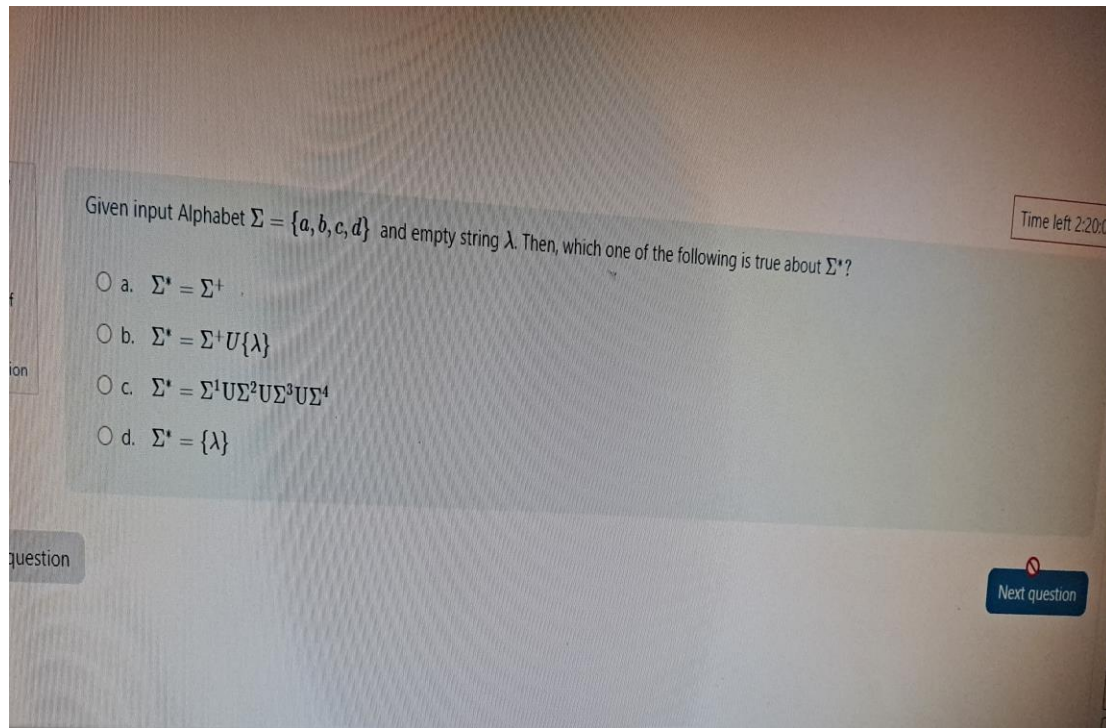
C. It is a temporary read problem

D. It is an incorrect summary problem

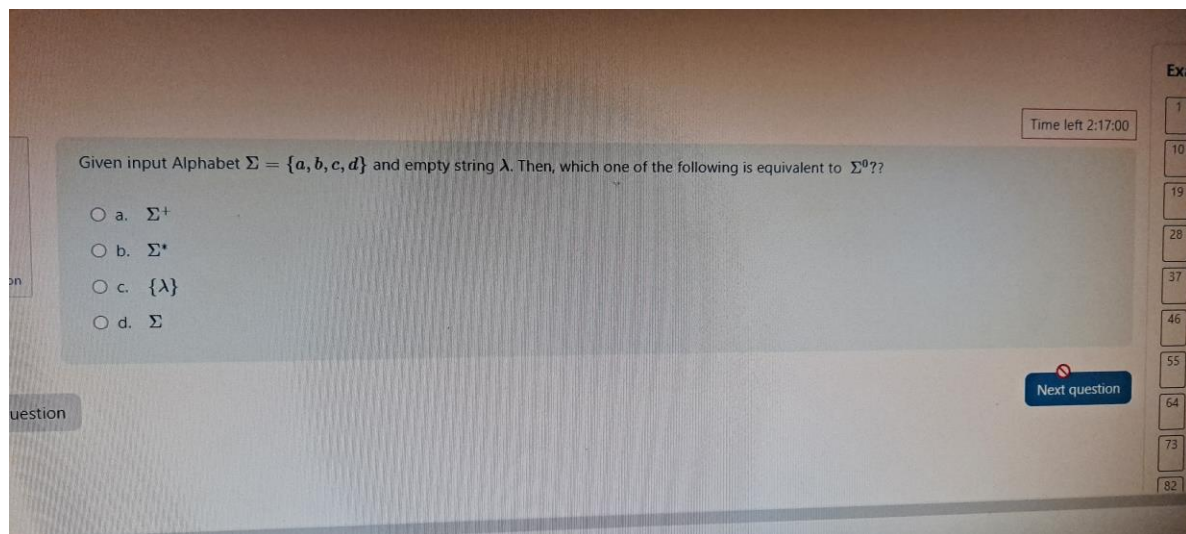
96.



97.



98.



99.

Back

Question 62

Not yet
answered

Marked out of
1.00

Flag question

Time left 2:16:49

Which one of the following is not delimiter of PHP code?

- ☐ a. `<script language="PHP"> ..... </script>`
- ☐ b. `<?php ..... ?>`
- ☐ c. `<caption> ..... </caption>`
- ☐ d. `<? ..... ?>`

Previous question

Next question

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Question 87
Not yet answered
Marked out of 1.00
Flag question

Time left 2:12:32

What is the output of the following C++ fragment code?

```
int a, b;  
a = 13;  
b = 9;  
while( (20 && 0) && (a > b) )  
{  
    cout << "Plants are our life";  
}
```

☐ a. It displays message "Plants are Our Life" 4 times

☐ b. It displays message "Plants are our life" infinitely

☐ c. It displays message "Plants are our life" 20 times

☐ d. No message will be displayed

Next question

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Time left 2:14:39

What is the output of the following C++ fragment code?

```
int a = 6, b = 8;  
int x = 2, y = 4;  
int c = (x > y ? (a--, x) : (b--, y));  
cout << "a= " << a;  
cout << "    b= " << b;  
cout << "    c= " << c;
```

☐ a. a = 6 b = 7 c = 4

☐ b. a = 6 b = 8 c = 4

☐ c. a = 5 b = 7 c = 2

☐ d. a = 5 b = 8 c = 4

question

Next question

Exam Navig

1	2	3
10	11	12
19	20	21
28	29	30
37	38	39
46	47	48
55	56	57
64	65	66
73	74	75
82	83	84
91	92	93
100		

101.

Question 91

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What is the output of the following Java fragment code?

```
int[] list = new int[4];  
int sum = 0;  
for(int i = 0; i < list.length; i++)  
{  
    list[i] = i*3;  
    sum += list[i];  
}  
System.out.print(sum);
```

- ☐ a. 18
- ☐ b. 6
- ☐ c. 3
- ☐ d. 36

Previous question

Time left 2:11:58

Next question

Exam

1	
10	
19	
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